

Week 3: Rules of inference

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- Valid arguments, premises, and conclusions
- (A lot of) rules of inference

1.6 Rules of Inference

Definition

- An **argument** in propositional logic is a compound proposition of the form

$$A : (P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q$$

A can have different truth values, depending on $P_1, \dots, P_n$, and Q .
where all P_i and Q can also be compound propositions.

- We call P_i **premises** and Q the **conclusion**.
assumptions

P1: Today it is sunny

P2: Today we have no class

Q: we go to barbecue.

A: $(P_1 \text{ and } P_2) \rightarrow Q$ is an argument.

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- An argument A is **valid** if A is a tautology.
correct way of reasoning (even though P_i or Q may not be correct)

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where all P_i and Q can also be compound propositions.

- We call P_i **premises** and Q the **conclusion**.
- An argument A is **valid** if A is a tautology.
- In contrast, if A is not a tautology, then we call A an **incorrect reasoning**, or a **fallacy**. at least in one case A is F

Modus ponens/the law of detachment.

$$\frac{((\underbrace{p \rightarrow q}_{P1} \wedge \underbrace{p}_{P2}) \rightarrow \underbrace{q}_{Q})}{\text{is a tautology.}}$$

p= P2	Q = q	P1: p→q	P1 and P2	Q= q	A: (P1 and P2) → Q
T	T	T	T	T	T
T	F	F	F	F	T
F	T	T	F	T	T
F	F	T	F	F	T

All statement within A, but not A itself, can be T/F, but A is always T.

That means using A as a way of thinking is correct.

It does not mean that individual or incomplete statements (p, q, p→q, ..) are T.

Modus ponens/the law of detachment.

$$((p \rightarrow q) \wedge p) \rightarrow q$$

is a tautology.

Meaning.

If “p implies q” holds, and p happens, then q follows.

Example.

$p \rightarrow q$: “If you have the password,
then you can log onto the network.”

p : “You have the password.”

$\therefore q$: “You can log onto the network.”

Non-example. Incorrect reasoning/fallacy

$$((p \rightarrow q) \wedge q) \rightarrow p$$

is not a tautology. (Try $p = F$ and $q = T$.)

$$A = ((F \rightarrow T) \text{ and } T) \rightarrow F = ((T \text{ and } T) \rightarrow F) = (T \rightarrow F) = F.$$

Non-example. Incorrect reasoning/fallacy

$$((p \rightarrow q) \wedge q) \rightarrow p$$

is not a tautology. (Try $p = F$ and $q = T$.)

Meaning. If “p implies q” holds, and q happens, it does not mean p also happens!!

Remember

You cannot use the conclusion to claim the assumption!!

Non-example. Incorrect reasoning/fallacy

$$((p \rightarrow q) \wedge q) \rightarrow p$$

is not a tautology.

Example (of this non-example).

$p \rightarrow q$: “If you have the password,
then you can log onto the network.”

q : “You have logged onto the network.”

does not imply p : “You have the password.”

(May be you hacked onto the network!!)

The meaning of valid arguments.

“ $A : (P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q$ ” is true

happens only in these cases.

Case 1. All P_i are true, and Q is true.

This means if all premises are true, then the conclusion must be true.

Case 2. At least one P_i is false, and Q can be either true or false.

In case 2. If one of these P_i is F, then $(P_1 \wedge P_2 \wedge \cdots \wedge P_n)$ is F, and
“ $(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q$ ” = “ $F \rightarrow Q$ ” is always T

Think about: if you don't eat (P), then you die (Q).

If you eat (P is F), then you cannot conclude you die or not.

The meaning of valid arguments.

“ $A : (P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q$ ” is true

happens only in these cases.

Case 1. All P_i are true, and Q is true.

This means if all premises are true, then the conclusion must be true.

Case 2. At least one P_i is false, and Q can be either true or false.

This means if at least one premise is false, then we cannot draw any conclusion.

Example. Suspects v. Criminals

Observation. B is murdered at home, and A is B 's spouse. A is a **suspect**, What do we need to convict A as a **murderer**?

witness	weapon	motivation	guilty
T	T	T	T
F	T	T	T/F no conclusion not enough evidence
T	F	T	T/F no conclusion not enough evidence
T	T	F	T/F
Any suspect is innocent until proven guilty (need all evidence)			

Other rules of inference

Modus tollens.

$$\text{A: } \frac{\frac{(p \rightarrow q)}{P1} \wedge \frac{\neg q}{P2}}{Q} \rightarrow \frac{\neg p}{Q}$$

is a tautology, that is, a valid argument.

This is another form of modus ponens:

$$\frac{((\neg q \rightarrow \neg p) \wedge \neg q)}{P1} \rightarrow \neg p$$

In P1, $(\neg q \rightarrow \neg p)$ is the contrapositive of $(p \rightarrow q)$, they are equivalent.

Other rules of inference

Modus tollens.

$$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$$

is a tautology, that is, a valid argument.

Meaning. If “p implies q” holds, but q does not happen, then p cannot happen neither.

Example.

$p \rightarrow q$: “If you have the password,
then you can log onto the network.”

$\neg q$: “You cannot log onto the network.”

$\therefore \neg p$: “You don’t have the password.”

Hypothetical syllogism

$$A: \underbrace{(p \rightarrow q)}_{P1} \wedge \underbrace{(q \rightarrow r)}_{P2} \rightarrow \underbrace{(p \rightarrow r)}_Q$$

is a tautology, that is, a valid argument.

The necessary condition of P1 is the same as the sufficient condition of P2, which is q.

Hypothetical syllogism

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

is a tautology, that is, a valid argument. **transitivity of implications**

Meaning. If “p implies q” and “q implies r”, then “p implies r”.

Example.

p : “It is raining today.”

q : “We don’t have a barbecue today.”

r : “We will have a barbecue tomorrow.”

conclude: $p \rightarrow r$: If it is raining today, then we go barbecue tomorrow.

Disjunctive syllogism

$$\frac{((\underbrace{p \vee q}_{P1}) \wedge \underbrace{\neg p}_{P2})}{\underbrace{q}_{Q}}$$

is a tautology, that is, a valid argument.

This is again another form of Modus ponens:

$$\frac{((\underbrace{\neg p \rightarrow q}_{P1}) \wedge \neg p)}{q}$$

Remember $(p \rightarrow q)$ is $(\neg p \text{ or } q)$, so in P1, $(p \rightarrow q)$ is $(\neg(\neg p) \text{ or } q) = (p \text{ or } q)$

Disjunctive syllogism

$$((p \vee q) \wedge \neg p) \rightarrow q$$

is a tautology, that is, a valid argument.

Meaning. If either p or q must happen, but p does not happen, then q must happen.

Example.

$p \vee q$: “Students in this class are Math major or Phy major.”

$\neg p$: “You are not Math major.”

$\therefore$ **q: You are in Phy major.**

Some easy valid arguments.

Addition

$\underline{p} \rightarrow \underline{(p \vee q)}, \quad q \rightarrow (p \vee q).$
P1 Q
are valid arguments.

Some easy valid arguments.

Addition

$$p \rightarrow (p \vee q), \quad q \rightarrow (p \vee q).$$

are valid arguments.

Simplification

$$\frac{(p \wedge q)}{P1} \rightarrow \frac{p}{Q}, \quad (p \wedge q) \rightarrow q.$$

are valid arguments.

Some easy valid arguments.

Addition

$$p \rightarrow (p \vee q), \quad q \rightarrow (p \vee q).$$

are valid arguments.

Simplification

$$(p \wedge q) \rightarrow p, \quad (p \wedge q) \rightarrow q.$$

are valid arguments.

Conjunction

$$\underbrace{((p) \wedge (q))}_{\text{P1} \quad \text{P2}} \rightarrow \underbrace{(p \wedge q)}_{\text{Q}}.$$

is a valid argument.

Example. Show that the premises $\underbrace{(p \wedge q) \vee r}_{P1}$ and $\underbrace{r \rightarrow s}_{P2}$ imply the conclusion $\underbrace{p \vee s}_{Q}$.

The argument is $A : ((P1 \wedge P2) \rightarrow Q) = (((p \wedge q) \vee r) \wedge (r \rightarrow s)) \rightarrow (p \vee s)$

Instead of using truth tables (that would be a 16-case table),
better use logical equivalence/rules of inference (valid arguments).

(observe that q and r do not appear in s, so think of the ways we can eliminate q and r)

Start from	$P1 \wedge P2 = ((p \wedge q) \vee r) \wedge (r \rightarrow s)$	
	$= (\sim (\sim(p \wedge q)) \vee r) \wedge (r \rightarrow s)$	Using $(a \rightarrow b) = (\sim a \vee b)$
	$= (\sim(p \wedge q) \rightarrow r) \wedge (r \rightarrow s)$	where $a = \sim(p \wedge q)$ and $b = r$.
	$\rightarrow (\sim(p \wedge q) \rightarrow s)$	(using hypothetical syllogism to eliminate r)
	$= ((p \wedge q) \vee s)$	Using $(a \rightarrow b) = (\sim a \vee b)$ again
	$= (p \vee s) \wedge (q \vee s)$	distributive law
	$\rightarrow (p \vee s) = Q$	(eliminate $(q \vee s)$ by simplification)

Universal instantiation

$$\forall x, P(x) \rightarrow \exists x, P(x)$$

is a valid argument. When $D = \{1,2\}$, then LHS = (P1 and P2), RHS = (P1 or P2)
This is coming from the calculation
(P1 and P2) $\rightarrow$ P1 (by simplification) $\rightarrow$ (P1 or P2) (by addition)

Universal generalization. Let $\mathcal{D}$ be the domain, then

$$P(c) \text{ for any } c \text{ in } \mathcal{D} \rightarrow \forall x, P(x)$$

is a valid argument.

- We cannot make any other assumptions about c (other than it comes from $\mathcal{D}$).

Universal modus ponens

$$\boxed{\forall c} \quad \frac{\underbrace{(\forall x(p(x) \rightarrow q(x)))}_{P1} \wedge \underbrace{p(c)}_{P2}}{A(c)} \rightarrow \underbrace{q(c)}_Q$$

is a valid argument.

Example.

The whole box is A(c)

“Everyone in this discrete mathematics class
has taken a course in computer science.”

“Marla is a student in this class.”

“Marla has taken a course in computer science.”

c = Marla

This argument applies to other choice c of students, as long as c is in $D = \{\text{persons}\}$
If c = Peter, then we can repeat the above arguments by replacing Marla by Peter.

Existential instantiation

$$\exists x, P(x) \rightarrow P(c) \text{ for some } c \text{ (in } \mathcal{D}\text{)} \\ \text{there exists}$$

is a valid argument.

- We cannot select an arbitrary value of c in $\mathcal{D}$.
That means there is a special (not arbitrary) c in $\mathcal{D}$ that satisfies $P(c)$.
- Usually we have no knowledge of c (other than its existence).

Existential generalization

$$P(c) \text{ for some } c \rightarrow \exists x, P(x)$$

is a valid argument.